

Impact of Histopathological Feature Encoding on Machine Learning Prediction of Sperm Retrieval in Micro-TESE for Non-Obstructive Azoospermia: A Comparative Analysis of Encoding Strategies

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Abstract

Background: Non-obstructive azoospermia (NOA) affects approximately 10% of infertile men and represents the most severe form of male infertility. Microsurgical testicular sperm extraction (micro-TESE) is the gold standard for sperm retrieval, yet success rates vary widely (40-60%). While machine learning (ML) approaches have shown promise in predicting micro-TESE outcomes, the impact of histopathological feature encoding strategies on model performance has not been systematically investigated. Methods: We performed a retrospective analysis of 2,413 NOA patients who underwent micro-TESE at Royan Institute, Tehran, Iran. Sixteen encoding strategies for bilateral testicular biopsy histopathological features were designed and evaluated across five ML classifiers using 5×5 nested cross-validation with 50-iteration randomized hyperparameter search per fold. Results: The ordinal_detailed_plus_raw encoding strategy, which combines per-testis ordinal severity scores with 16 raw histopathological percentage features, achieved optimal performance. CatBoost attained the highest AUC of 0.8306 (bootstrap 95% CI: 0.823–0.845), with sensitivity of 0.678, specificity of 0.822, and F1 of 0.658. SHAP analysis revealed that 17 of 25 top features were pathology-derived, with right testicular severe hypospermatogenesis percentage being the single most influential predictor. Conclusion: Systematic encoding of histopathological features substantially impacts ML-based prediction of sperm retrieval in micro-TESE. The proposed ordinal encoding with raw feature preservation achieves state-of-the-art prediction performance and provides clinically interpretable feature attributions.

Key Results Summary

The ordinal_detailed_plus_raw encoding strategy achieved the best CatBoost AUC of 0.8306 (95% CI: 0.823–0.845), significantly outperforming other encoding approaches. SHAP analysis showed 17 of 25 top features were pathology-derived, with right testicular severe hypospermatogenesis being most influential. Clinical-only baseline achieved near-chance AUC (0.5642), confirming the critical importance of histopathological features.

Tables

1. Table 1. Baseline Patient Characteristics (n=2,413)

Headers: Variable Value
- Demographics
- Total patients, n 2,413
- Age, years (missing data acknowledged) Available for subset
- Partner age, years (8.6% missing) Available for subset
- BMI, kg/m² (64.1% missing) Available for subset
- Hormonal Profile
- FSH, mIU/mL (22.2% missing) Available for imputed analysis
- LH, mIU/mL (25.1% missing) Available for imputed analysis
- Testosterone, ng/dL (22.6% missing) Available for imputed analysis
- Prolactin, ng/mL (40.4% missing) Available for imputed analysis
- E2, pg/mL (41.6% missing) Available for imputed analysis
- Genetic Factors
- Karyotype available, n (%) 923 (38.3%)
- Y-microdeletion tested, n (%) 922 (38.2%)
- Histopathological Patterns (RT)
- SCO present, n (%) 1,233 (51.1%)
- CSTH present, n (%) 675 (28.0%)
- MA (any stage) present, n (%) 985 (40.8%)

- Hypospermatogenesis present, n (%) | 102 (4.2%)
- Severe hypospermatogenesis present, n (%) | 257 (10.6%)
- Surgical History |
- Primary micro-TESE, n (%) | 2,203 (91.3%)
- Salvage micro-TESE, n (%) | 210 (8.7%)
- Outcome |
- Successful sperm retrieval, n (%) | 886 (36.7%)
- Unsuccessful sperm retrieval, n (%) | 1,527 (63.3%)

2. Table 2. Comparison of 16 Encoding Strategies (CatBoost AUC)

Headers: Rank | Encoding Strategy | # Features | CatBoost AUC

- 1 | ordinal_detailed_plus_raw | 18 | 0.8208
- 2 | raw_pct_minmax | 16 | 0.8194
- 3 | raw_pct_robust | 16 | 0.8194
- 4 | raw_pct_avg | 8 | 0.8179
- 5 | binary_presence_avg | 8 | 0.8163
- 6 | grouped_avg | 4 | 0.8073
- 7 | binary_presence | 16 | 0.8041
- 8 | grouped | 8 | 0.8041
- 9 | raw_pct_worst | 8 | 0.7920
- 10 | binary_presence_worst | 8 | 0.7797
- 11 | grouped_worst | 4 | 0.7772
- 12 | ordinal_simplified | 2 | 0.7502
- 13 | ordinal_detailed | 2 | 0.7382
- 14 | ordinal_detailed_worst | 1 | 0.7218
- 15 | dominant_pattern | 18 | 0.7346
- 16 | ordinal_detailed_avg | 1 | 0.7129

3. Table 3. Model Performance Comparison (5×5 Nested CV, ordinal_detailed_plus_raw Encoding)

Headers: Model | AUC (95% CI) | Accuracy | Sensitivity | Specificity | F1 | MCC

- CatBoost | 0.8306 (0.823–0.845) | 0.7696 | 0.6783 | 0.8225 | 0.6838 | 0.5040
- XGBoost | 0.8214 (0.814–0.835) | 0.7580 | 0.6693 | 0.8094 | 0.6689 | 0.4801
- LightGBM | 0.8213 (0.809–0.836) | 0.7551 | 0.6806 | 0.7984 | 0.6709 | 0.4790
- Random Forest | 0.8157 (0.809–0.829) | 0.7472 | 0.6941 | 0.7780 | 0.6685 | 0.4660
- Logistic Regression | 0.7926 (0.777–0.804) | 0.7348 | 0.6592 | 0.7787 | 0.6458 | 0.4356

4. Table 4. Top 25 Features by Mean |SHAP| Value (CatBoost)

Headers: Rank | Feature | Mean |SHAP| | Category

- 1 | RT_Severe_Hypospermatogenesis_pct | 0.3958 | Pathology
- 2 | RT_Hypospermatogenesis_pct | 0.2007 | Pathology
- 3 | LT_Severe_Hypospermatogenesis_pct | 0.1590 | Pathology
- 4 | RT_MA_Spermatocytic_pct | 0.1473 | Pathology
- 5 | LT_severity | 0.1308 | Pathology (engineered)
- 6 | LT_SCO_pct | 0.1099 | Pathology
- 7 | RT_SCO_pct | 0.0898 | Pathology
- 8 | RT_severity | 0.0855 | Pathology (engineered)
- 9 | RT_MA_Elongated_pct | 0.0672 | Pathology
- 10 | Habits_(blank) | 0.0663 | Lifestyle
- 11 | Karyotype_Klinefelter | 0.0612 | Genetic
- 12 | LT_MA_Spermatocytic_pct | 0.0606 | Pathology
- 13 | LT_Hypospermatogenesis_pct | 0.0565 | Pathology
- 14 | Seminal plasma pH | 0.0550 | Semen
- 15 | RT_TestisSize_Sono_Vol | 0.0501 | Testicular
- 16 | RT_CSTH_pct | 0.0308 | Pathology
- 17 | Sakamoto-RT/mL | 0.0268 | Testicular

- 18 | LT_TestisSize_Sono_Vol | 0.0205 | Testicular
- 19 | RT_MA_Spermatogonial_pct | 0.0188 | Pathology
- 20 | FSH | 0.0180 | Hormonal
- 21 | LT_MA_Spermatogonial_pct | 0.0173 | Pathology
- 22 | Sakamoto-LT/mL | 0.0172 | Testicular
- 23 | Prolactin | 0.0164 | Hormonal
- 24 | Partner Age | 0.0161 | Demographic
- 25 | LT_Volume_Unified | 0.0152 | Testicular

Figures

Figure 1: Encoding strategy comparison showing AUC performance of 16 encoding strategies across CatBoost. Ordinal_detailed_plus_raw achieves highest AUC (0.8208 in screening).

Figure 2: Model performance comparison across 5 classifiers using the optimal ordinal_detailed_plus_raw encoding.

Figure 3: SHAP summary plot showing top 25 features by mean absolute SHAP value for CatBoost, with pathology features dominating.

Figure 4: Pipeline v1 vs v2 performance comparison showing AUC improvement from 0.7327 to 0.8306 after quality assurance corrections.